Sampling complexity of the Random Hierarchy Model under the LS (local score) machine

There is no finite P. Sampling complexity is exp(Theta(n)) in the number of sites -- the LS machine gains nothing from the hierarchy.

Criterion (hshunt): a generated grid counts only if it is NOT in the training set AND fully grammatical.

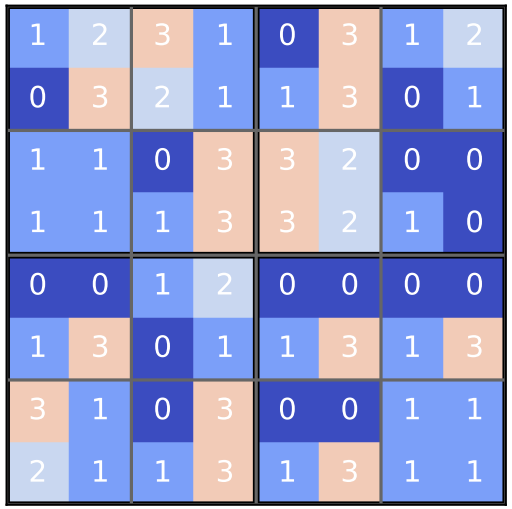
Because every training grid is grammatical, success = grammatical + novel - 1, so the question is only whether grammaticality ever beats the memorization rate. Over k in {3,5,7,15}, v in {2,3,4,6,8}, m in {2..6}, L in {1,2,3} and P from 10 to 300,000, it never does by more than a margin that vanishes exponentially in the number of sites.

Two legs: a window of side k can police the grammar only up to block side ~k/2, which caps grammaticality at a value independent of P (derivation, exact); and widening the window costs data exponentially in its area (measured).

Repo convolutional_diffusion @ c86b9f5; rule seeds 1000-1002; alpha = 1e-3; one site revealed at a time (exact ancestral sampling); n_gen 512-8192.

Tokenization: what a "sample" is, and what the window sees

one sample: 8x8 grid of tokens in {0..3}



Tokens. Every site holds one of v = 4 symbols. There is no pixel intensity and no ordering: token 3 is not "between" 2 and 4. A grid is 8x8 = 64 categorical sites.

Grammar. A root symbol expands 3 times. Each symbol is replaced by one of m = 3 synonymous 2x2 "words", drawn once and then fixed. So the grey lines at left bound aligned 2x2 blocks (one level-3 symbol each) and the black lines bound aligned 4x4 blocks (one level-2 symbol each). A grid is grammatical iff some root derives it.

Diffusion. Absorbing-state ("erasure"): each site is independently replaced by a MASK token with probability p(t). Unmasked sites are never corrupted, so exact Bayes denoising is patch matching -- no score approximation enters.

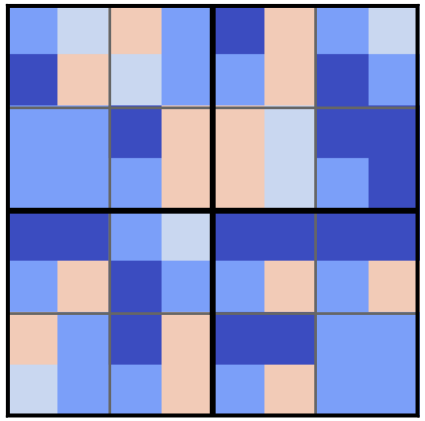
LS machine. The posterior at a masked site is the empirical distribution of the centre tokens of all training k x k patches that agree with the observed sites inside that window, with Laplace smoothing alpha. Sites are revealed one at a time (exact ancestral sampling; revealing in parallel from marginals adds ~n^2/2T independence errors, which alone made 12% of grids ungrammatical).

Success (hshunt's criterion). A generated grid counts only if it is NOT in the training set AND parses completely. Since every training grid is grammatical,

$$\text{success} = \text{grammatical} + \text{novel} - 1$$

Example traces: what the machine actually emits, parsed level by level

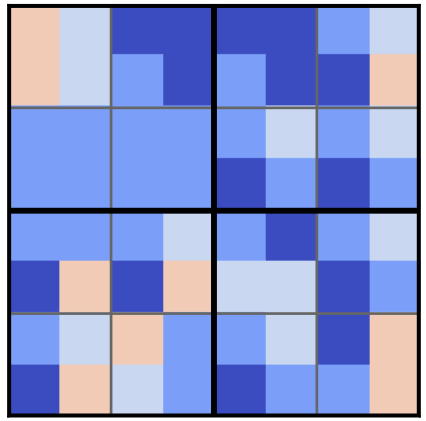
a true RHM sample



```
parse, bottom up
(mean of 6 draws):
  OK leaves      1.00
  OK 2x2 blocks  1.00
  OK 4x4 blocks  1.00
  OK whole grid  1.00

grammatical by construction.
```

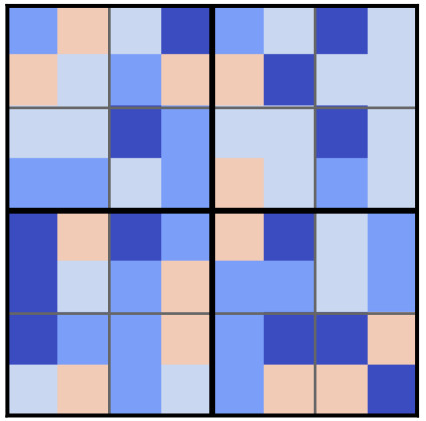
LS machine, k=3
(P=600)



```
parse, bottom up
(mean of 6 draws):
  OK leaves      1.00
  XX 2x2 blocks  0.95
  XX 4x4 blocks  0.25
  XX whole grid  0.00

window reaches 1 site:
seals 2x2 blocks,
blind to 4x4 and up.
novel 1.00
```

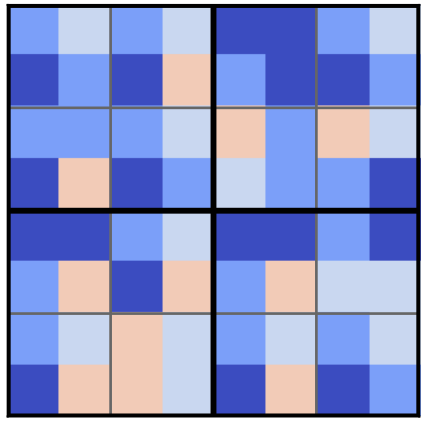
LS machine, k=7
(P=600)



```
parse, bottom up
(mean of 6 draws):
  OK leaves      1.00
  XX 2x2 blocks  0.07
  XX 4x4 blocks  0.00
  XX whole grid  0.00

window reaches 3:
seals 4x4 too, but N_7
is far too big to fill.
novel 1.00
```

LS machine, k=15
(P=600)



```
parse, bottom up
(mean of 6 draws):
  OK leaves      1.00
  OK 2x2 blocks  1.00
  OK 4x4 blocks  1.00
  OK whole grid  1.00

window = whole grid:
perfect grammar,
but it is a copy.
novel 0.00
```

Law 1 (derivation): the window sets a hard ceiling, independent of P

A k x k window centred on a site reaches only k//2, so it contains the aligned block of side b through that site only when k >= 2b - 1. Write

$$j^* = \text{floor}(\log_s((k+1)/2))$$

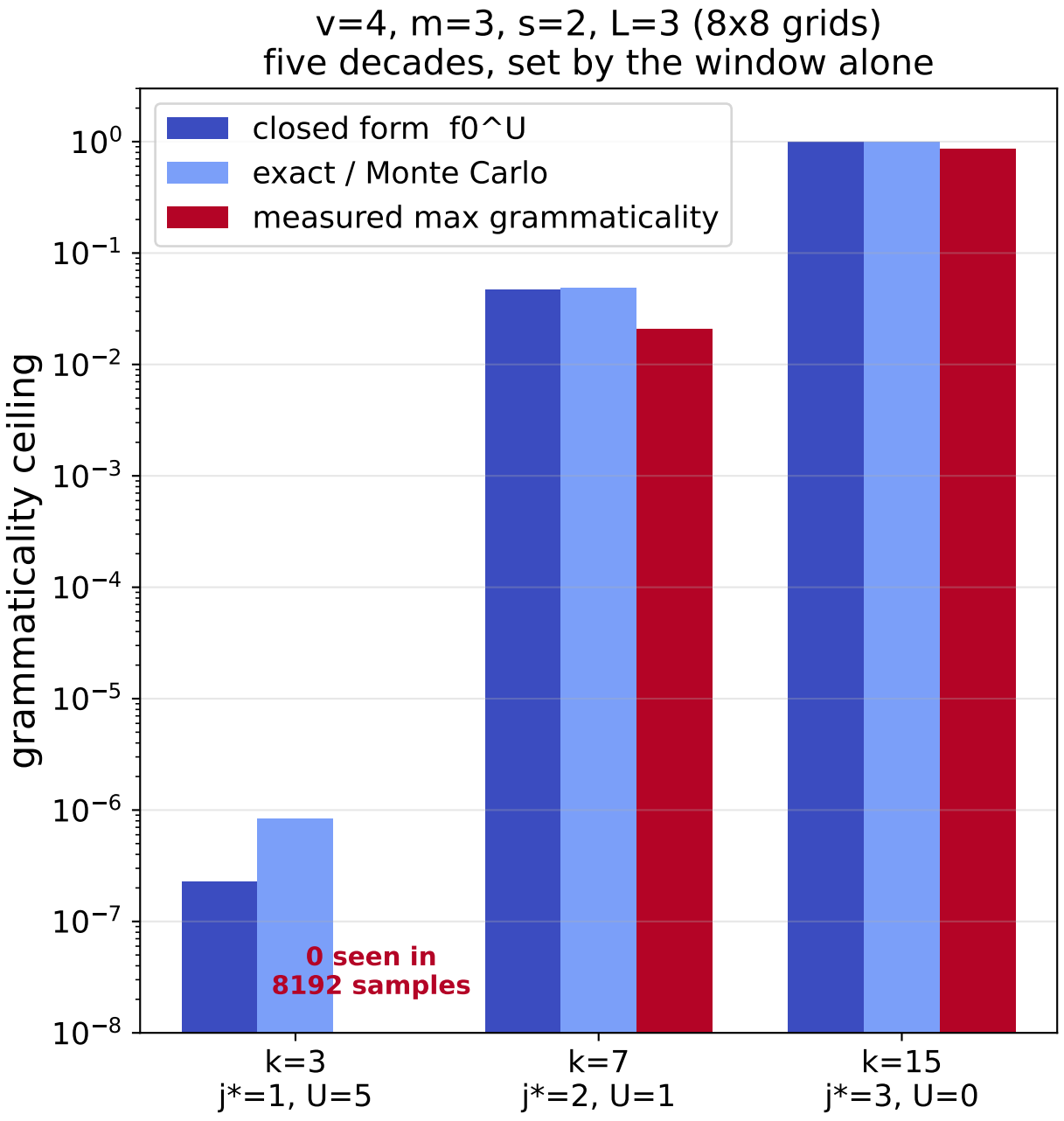
for the deepest level it can police. With unlimited data it then samples the locally-valid set $R_{j^*}^{((d/s^{j^*})^2)}$, of which the grammar is a fraction

$$C(k) = |R_L| \,/\, |R_{j^*}|^{(s^{2(L-j^*)})} \quad \text{(exact)}$$

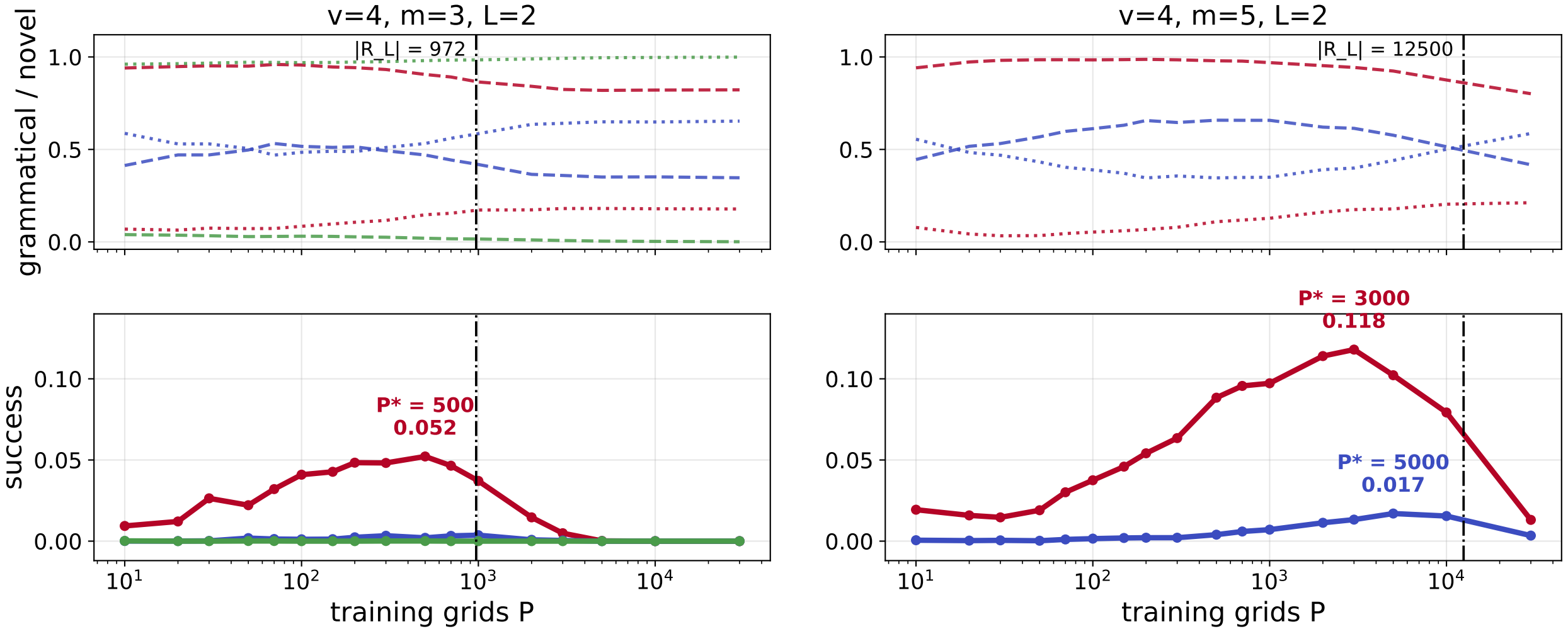
with R_j = distinct leaf renderings of an aligned s^j block. With no word collisions $|R_j| = v\, m^{((s^{2j}-1)/(s^2-1))}$, and this collapses algebraically to

$$C(k) = f0^U, \quad f0 = m \,/\, v^{(s^2-1)}, \quad U = (s^{2(L-j^*)}-1)/(s^2-1)$$

U counts the rule instances above the window's reach. Over the 150 enumerable L=2 cases the closed form is exact in exactly the 70 drawn without a word collision, and wrong in exactly the 80 with one -- no exceptions. There the counting form is the correct one. Collisions renormalize f0; they do not touch U.

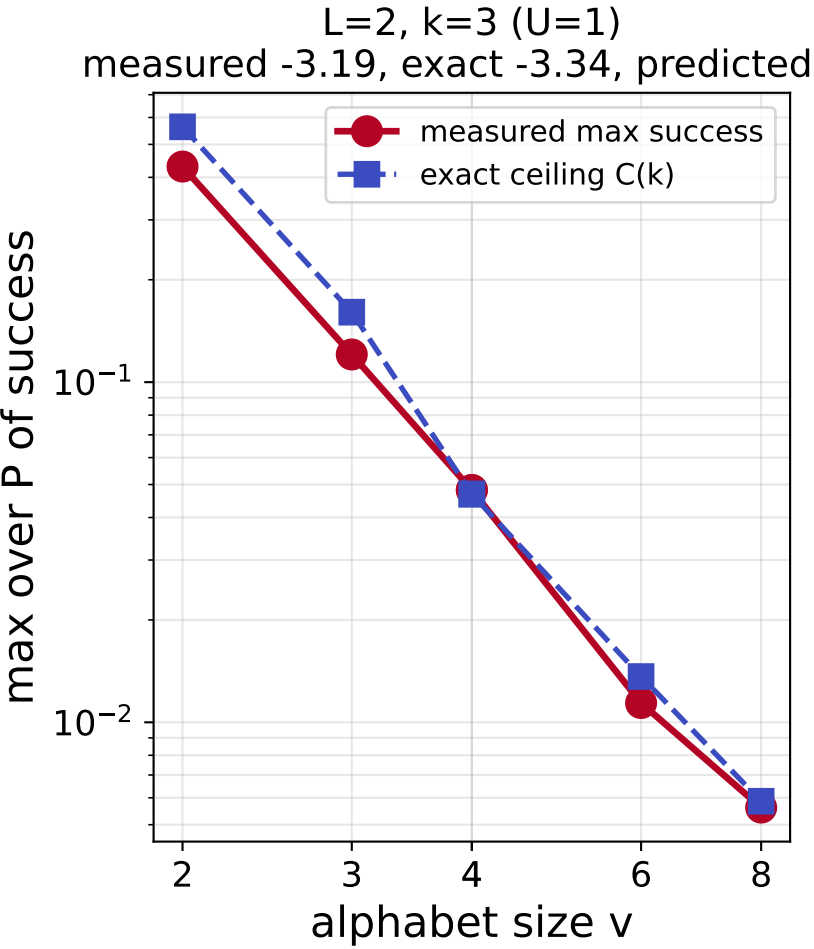


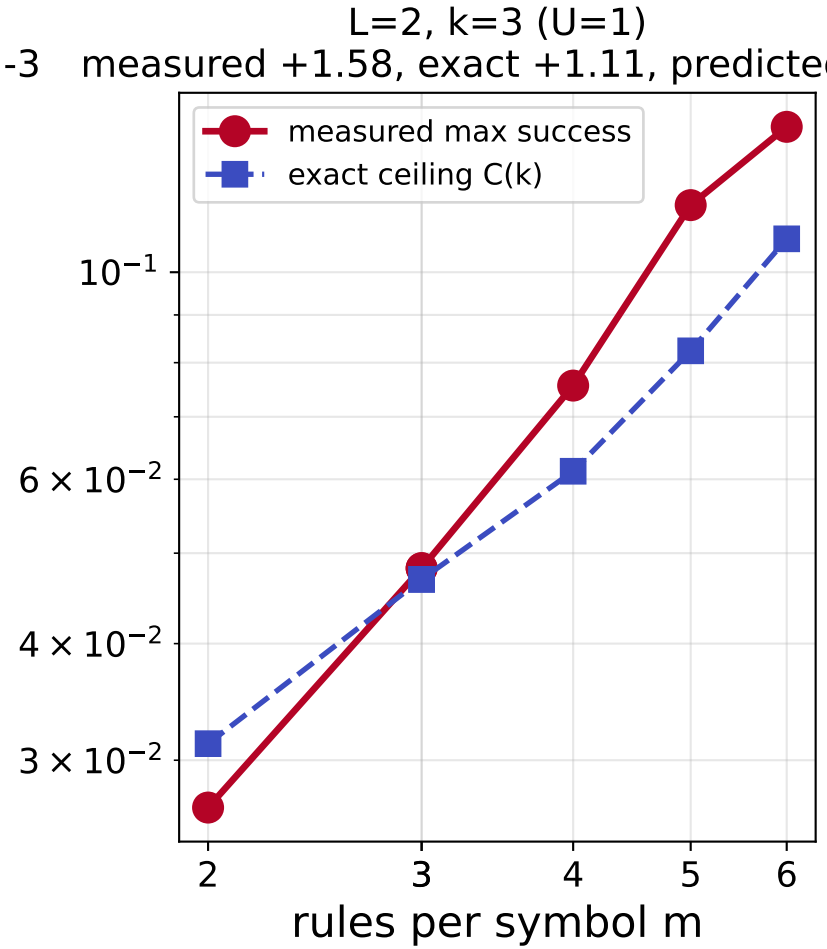
Law 2 (measurement): success peaks at a finite P, then more data destroys it

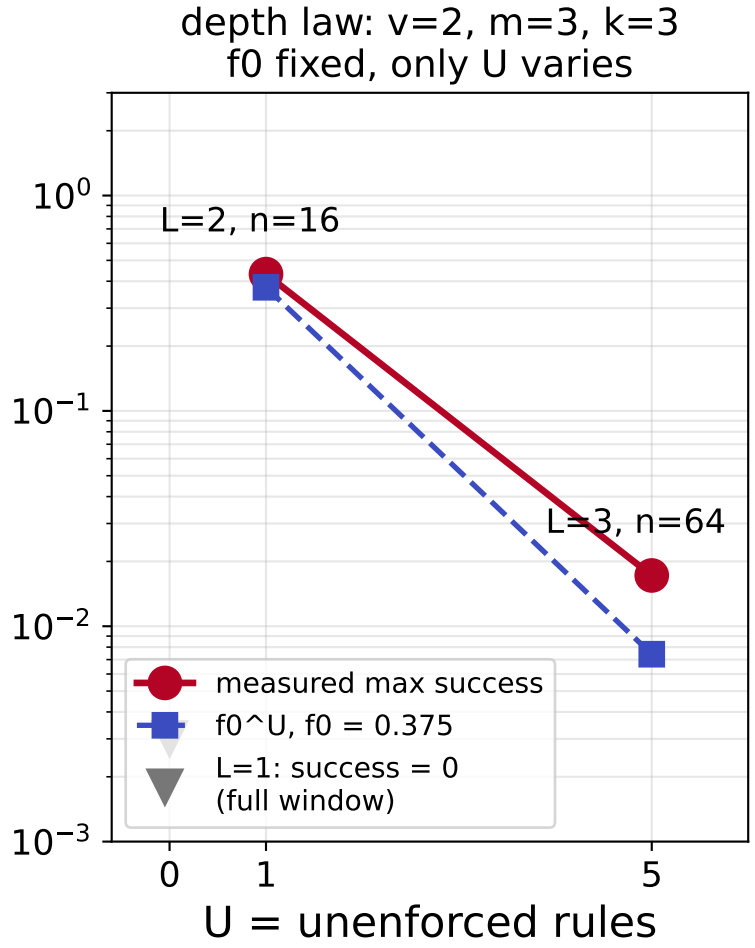


Dash-dot = size of the whole grammar. k=7 = 2d-1 is the full window:
grammaticality 0.999, novelty 0.0009; success 1.6e-4 at P=10, exactly 0 at the other 16 P.

The exponents: success attains the derived ceiling

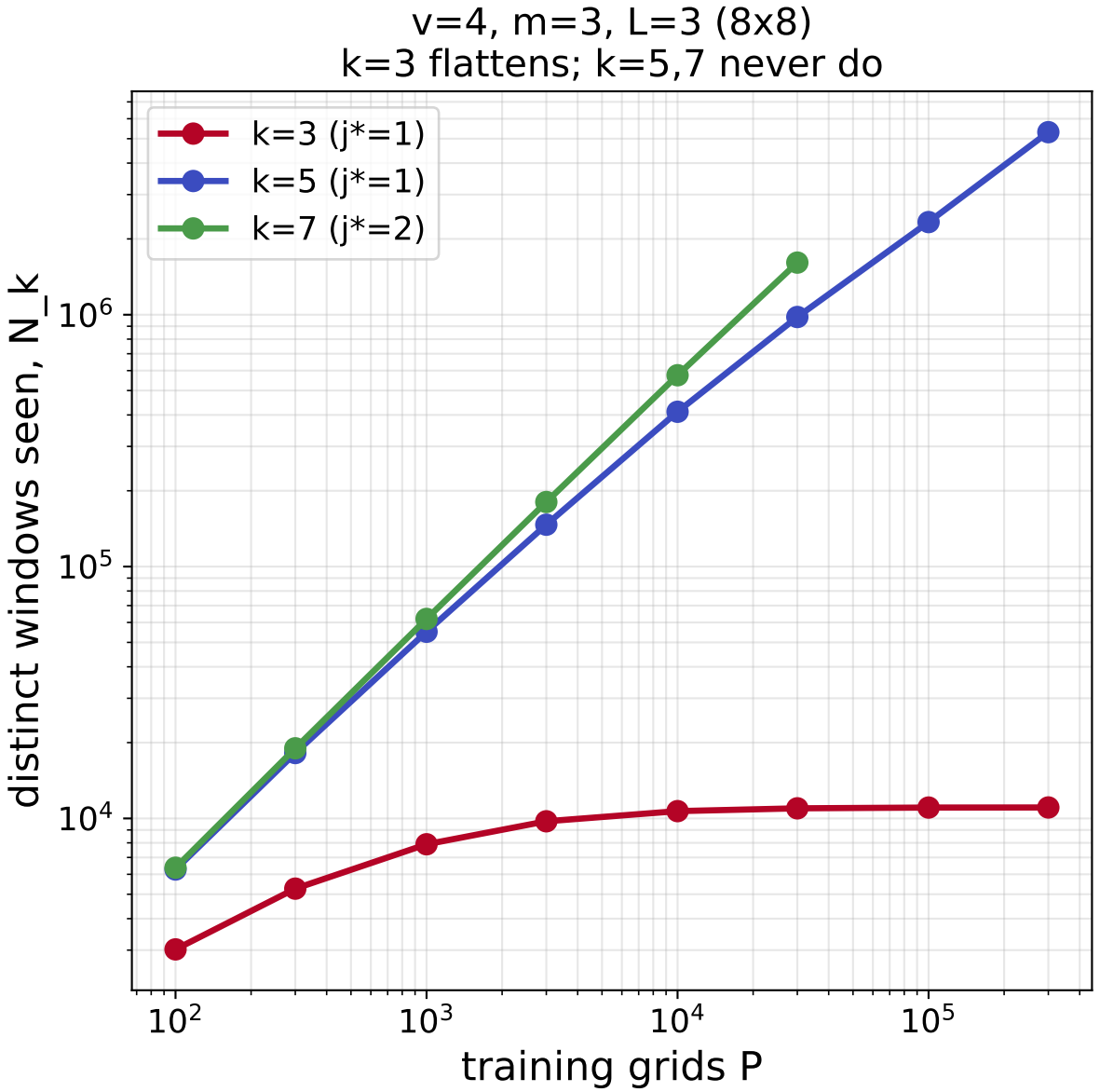






The v and m panels measure the ceiling exponents directly: success tracks the exactly-computed C(k) across two decades.
L=1 is excluded from the depth trend because there k=3 = 2d-1 is already the full window, so grammaticality is 1 and novelty 0.
Both ends of the k ladder give success = 0, for opposite reasons.

Law 3: the data cost of a larger window, N_k = distinct k x k windows



L = 3, at P = 3e5 (k=3,5) / 3e4 (k=7)

v	m	k	N_k	x/decade	verdict
2	3	3	428	1.00	flat
3	3	3	3576	1.01	flat
4	2	3	3389	1.00	flat
4	3	3	13602	1.02	flat
4	3	5	5318108	5.43	still climbing
4	3	7	1610432	8.92	still climbing
4	4	3	25681	1.03	flat
4	5	3	42359	1.02	flat
6	3	3	79974	1.16	still climbing
8	3	3	174247	1.32	still climbing

k=3 is within 3% per decade in 6 of the 8 grammars (v=6 1.16x, v=8 1.32x still climbing) -- a bigger polynomial, not another regime.
Across all 8, k=5 grows 1.80-7.83 per decade and k=7 6.78-8.92: neither ever saturates.

Fitted at k=3: N_3 ~ v^4.4 m^2.8 at L=3, v^2.1 m^3.7 at L=2 -- polynomial either way, which is why k=3 is the only window learnable at all at L=3.

A window of side k = 2b-1 is fixed by the 3x3 aligned b-blocks it touches, so N_k grows like m^Theta(k^2/(s^2-1)): EXPONENTIAL IN WINDOW AREA.

The scaling law

Combining the two legs. Success >= eps needs U = O(1), i.e. s^(2(L-j*)) = O(1),
i.e. the window must scale with the whole grid: k = Theta(d) = Theta(sqrt(n)).
But then the patch table has |R_j*| = v m^(Theta(n/(s^2-1))) entries, and P must fill
it while staying below the memorization point |R_L| -- and at k = 2d-1 those two
coincide exactly, so the interval where success > 0 closes.

P* = v . m^(Theta(n / (s^2 - 1))) = exp(Theta(n)), n = s^(2L) sites

At fixed window k: success ~ f0^((n/a - 1)/(s^2-1)), a = s^(2j*) ~ ((k+1)/2)^2
so ln(1/success) = Theta(n / k^2).

The LS machine extracts NO benefit from the hierarchy: its sampling complexity
is exponential in the number of sites, the same as memorizing the whole
distribution. Under the joint criterion (novel AND grammatical) there is no
finite P at all -- grammaticality reaches 1 only at the window size where
novelty is identically 0.

Previous best on this eval: none. This is the repo's first measurement of RHM
sampling complexity for the LS machine; the grammar oracle (rhm_grammar.py) and
the patch-count machine were written for it.

Why this matters for MLS: the obstruction is compositional, not statistical.
One window cannot police two scales at once. A hierarchy of windows -- a 3x3
machine whose output feeds a second 3x3 machine on inferred level-1 symbols --
would police scale s^2 at the cost of two tables of size |R_1|: POLYNOMIAL
instead of exponential. That is the natural next experiment.